

Spectral Quantification of MRSI Data Using Spatiospectral Constraints

Qiang Ning, Chao Ma, Fan Lam, Bryan Clifford, Zhi-Pei Liang

November 11, 2015

1 Synopsis

A new method is proposed for spectral quantitation of MRSI data. The method has two main features: 1) incorporation of prior spectral knowledge in the form of basis functions obtained by quantum simulation, and 2) incorporation of prior spatial knowledge by penalizing smoothness within each type of tissue. An efficient algorithm is also proposed to solve the underlying optimization problem, and its effectiveness for extracting quantitative spectral information from noisy MRSI data is demonstrated by comparing it with one of the state-of-the-art methods.

2 Introduction

Spectral quantification is a key step in practical applications of MRSI. Many methods have been proposed to solve this problem, which can be roughly classified into three groups: 1) basic frequency integration methods; 2) classical linear prediction methods exploiting the linear predictability of spectral signals [1, 2]; 3) modern methods incorporating strong spectral constraints in the form of basis functions [3, 4, 5]. In this work, we propose a new method that enables the use of both spectral and spatial constraints. In contrast to existing methods that perform quantification point-by-point independently, the proposed method solves the spectral quantification jointly, incorporating spatial prior knowledge to further improve the quantification performance [6].

3 Method

The proposed method has two main features: 1) the use of prior spectral information in the form of spectral basis functions, and 2) the use of prior spatial information in the form of spatial regularization. More specifically, we express the MRSI signal as $d(\mathbf{x}, t) =$

$\sum_{n=1}^N a_n(\mathbf{x}) e^{-\frac{t}{T_2^*(\mathbf{x})}} \varphi_n(t) + n(\mathbf{x}, t)$, where $\varphi_n(t)$ is the basis function for the n -th metabolite, and noise $n(\mathbf{x}, t)$ follows a white Gaussian distribution, or more compactly,

$$\mathbf{d} = \mathbf{K}(\mathbf{T}_2^*)\mathbf{a} + \mathbf{n},$$

where $\mathbf{d}$, $\mathbf{T}_2^*$, $\mathbf{a}$, and $\mathbf{n}$ are each discretized and flattened into a vector, and operator $\mathbf{K}$ represents the forward model dependent on $\mathbf{T}_2^*$. The model parameters are then determined by solving the following optimization problem,

$$(\hat{\mathbf{a}}, \hat{\mathbf{T}}_2^*) = \arg \min_{\mathbf{a}, \mathbf{T}_2^*} \|\mathbf{d} - \mathbf{K}(\mathbf{T}_2^*)\mathbf{a}\|_2^2 + \sum_{m=1}^M \sum_{n=1}^N \lambda_{m,n} \|\nabla \Omega_m \mathbf{a}^{(n)}\|_2^2 + \sum_{n=1}^N \mu_n \|\nabla \mathbf{a}^{(n)}\|_1,$$

where $\lambda_{m,n} > 0$ and $\mu_n > 0$ are regularization parameters, operator ∇ denotes the first order finite difference, Ω_m is the spatial support for the m -th tissue type, and the superscript "(n)" of $\mathbf{a}$ represents the n -th metabolite. The first regularization term penalizes the regional spatial variation (e.g., within gray matter, white matter, and CSF, respectively), and the second regularization term penalizes the global spatial variation (e.g., to avoid the artificial boundaries created by strict regional constraints). This formulation is motivated by the fact that in biological imaging samples (the human brain, calf muscles, etc.), the spatial distribution of a metabolite is usually similar within each type of tissue, but may also be very different across different tissue types (e.g., CSF is known to have negligible metabolites).

In practice, the spectral basis functions are readily available by either quantum mechanics simulation or in vitro experiments, and the spatial masks for different tissues can be obtained through segmentation of corresponding anatomical images. To solve the joint quantification problem efficiently, we can firstly solve the nonlinear parameters (i.e., $\mathbf{T}_2^*$) without incorporating spatial constraints using state-of-the-art methods, and then fix all the nonlinear parameters and solve the linear coefficients. If nonlinear terms are fixed, the resulting problem is an ℓ_1 regularized least-squares problem and can be solved using the alternating direction method for multipliers (ADMM) algorithm [7].

4 Results

The proposed method has been evaluated using both simulated and experimental data. The simulated MRSI dataset (Fig. 1a) was generated from a computational phantom which contains six common metabolites, NAA, Cre, Cho, Glu, mIn, and Gln. Metabolite distributions were designed such that they are smooth within each type of tissue, but different across different tissues (Fig. 1b). White Gaussian noise was also added to the dataset (Fig. 1c), with three typical spectra shown in Fig. 1d. In Fig. 2, the proposed quantification method is compared with one state-of-the-art method, QUEST [3], which performs quantification point-by-point independently. As can be seen, QUEST still had

large spatial variations, but the proposed method significantly improved the quantification and were much closer to the true distribution. Spectral estimation at some representative points are also evaluated as in Fig. 3. Both methods had residuals close to the noise floor (column 4 and 5), indicating comparable performance in terms of data fitting, but the spectral errors (difference between the true spectra) of QUEST were very large, which was reduced by the proposed method.

Experimental results from a healthy volunteer on a 3T MRI scanner are also shown in Fig. 4-5. The data were acquired using an echo-planar spectroscopic imaging (EPSI) sequence with bipolar acquisition. The estimated concentration maps and fitting results are compared with QUEST, which clearly show the performance improvement of the proposed method over QUEST.

5 Conclusion

A novel quantification method is proposed to address the spectral quantification problem for MRSI. In the proposed method, both spectral and spatial prior information is incorporated in a joint estimation framework to improve quantitation accuracy, and its effectiveness has been demonstrated via experimental results. The proposed method should prove useful in various quantitative MRSI studies.

6 Acknowledgements

This work was supported in part by the National Institutes of Health (NIH-1RO1- EB013695 and NIH-R21EB021013-01).

References

- [1] Barkhuijsen H, de Beer R, van Ormondt D, Improved algorithm for noniterative time-domain model fitting to exponentially damped magnetic resonance signals. *Journal of Magnetic Resonance* 1987;pp. 553–557.
- [2] Barkhuijsen H, de Beer R, Bovée WMMJ, van Ormondt D, Retrieval of frequencies, amplitudes, damping factors, and phases from time-domain signals using a linear least-squares procedure. *Journal of Magnetic Resonance* 1985;61:465–481.
- [3] Ratiney H, Sdika M, Coenradie Y, Cavassila S, van Ormondt D, Graveron-Demilly D, Time-domain semi-parametric estimation based on a metabolite basis set. *NMR in Biomedicine* 2005;18:1–13.

- [4] Young K, Govindaraju V, Soher BJ, Maudsley AA, Automated spectral analysis I: Formation of a priori information by spectral simulation. *Magnetic Resonance in Medicine* 1998;40:812–815.
- [5] Provencher SW, Estimation of metabolite concentrations from localized in vivo proton NMR spectra. *Magnetic Resonance in Medicine* 1993;30:672–679.
- [6] Ning Q, Ma C, Liang ZP, Spectral estimation for magnetic resonance spectroscopic imaging with spatial sparsity constraints. In: *Biomedical Imaging (ISBI), 2015 IEEE 12th International Symposium on*, IEEE, 2015; pp. 1482–1485.
- [7] Guo W, Qin J, Yin W, A new detail-preserving regularizaiton scheme. *SIAM Journal on Imaging Sciences* 2014;7:1309–1334.
- [8] Lam F, Liang ZP, A subspace approach to high-resolution spectroscopic imaging. *Magn Reson Med* 2014;71:1349 – 1357.

Figure 1: (a) Spectral basis functions of the six metabolites used to synthesize the MRSI dataset; (b) spectral integral of the synthesized dataset; (c) spectral integral after adding white Gaussian noise; and (d) representative spectra, with their locations indicated by the red dots in (c).

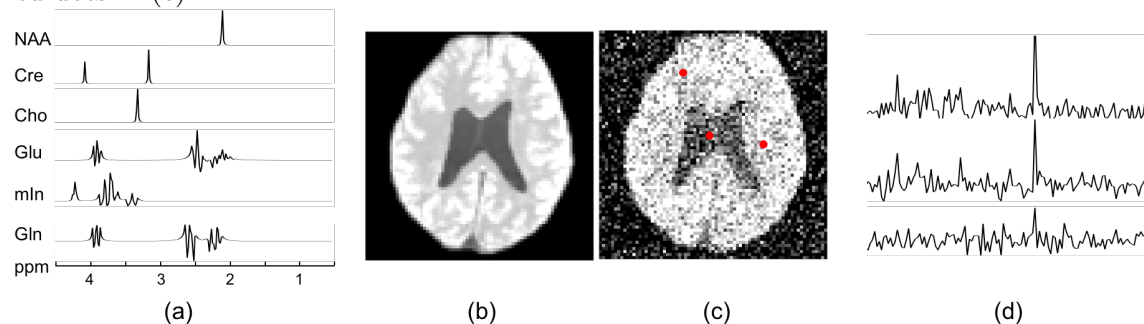


Figure 2: True metabolite distributions, metabolite distributions estimated by QUEST and by the proposed are shown in three rows, respectively. Because of the inherent difficulty in separating Glu from Gln, the conventional way is to show Glx (Glu+Gln) instead. Note the significant improvement of the proposed over QUEST.

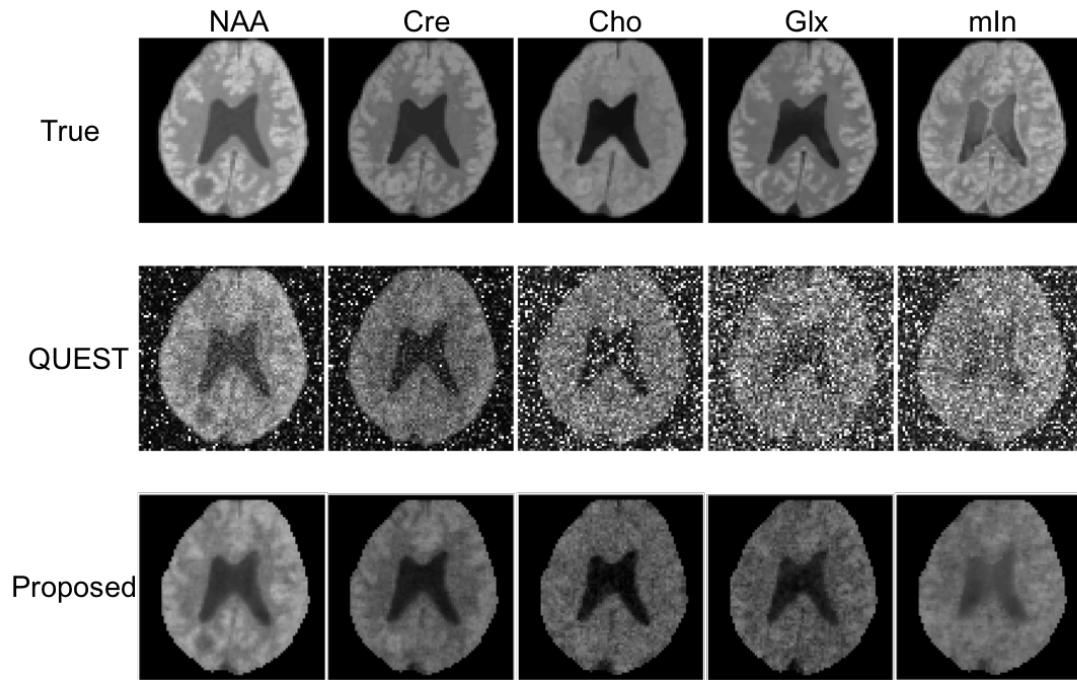


Figure 3: Spectral estimation results at some typical points are shown in (a)-(d). For both QUEST and the proposed, the difference between the true spectrum and the estimated spectrum is shown under the estimated spectrum. The right two columns represent the fitting residuals of two methods.

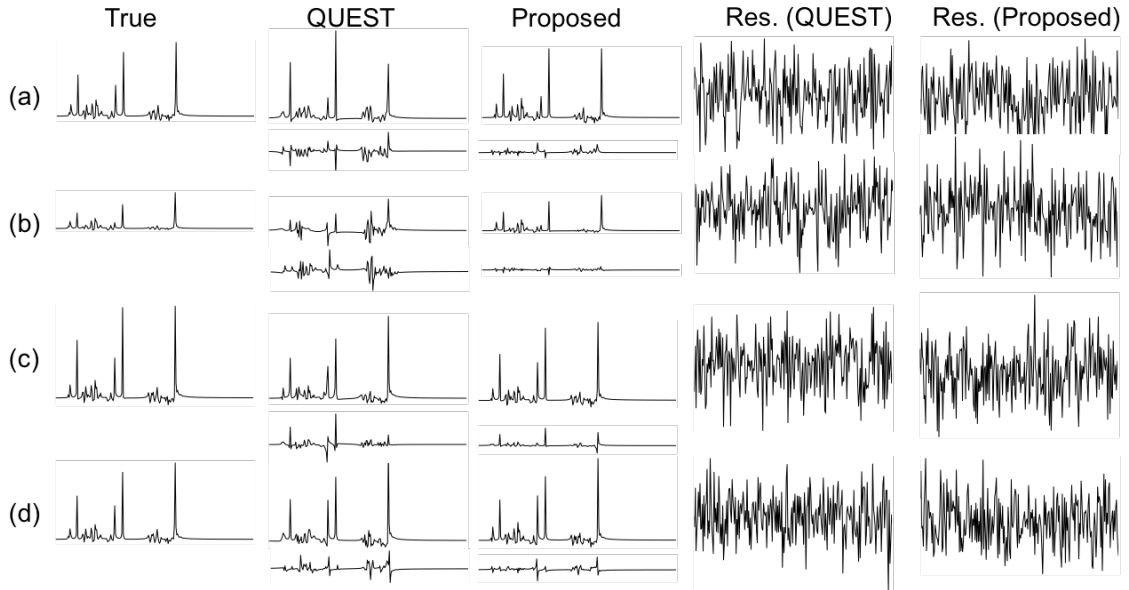


Figure 4: A comparison between the quantification results of QUEST and the proposed method is shown here. It can be observed that the proposed method reduced the spatial variation.

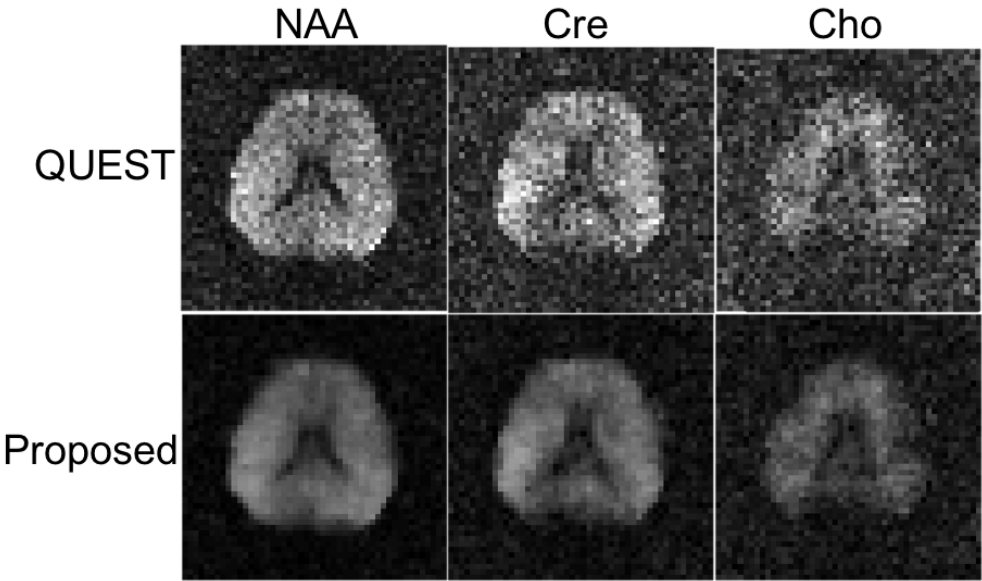


Figure 5: The spectral fitting results from a representative point are compared here. (a) fitting of QUEST; (b) fitting of the proposed ; (c) residual of QUEST; and (d) residual of the proposed. Note the similar fitting performance of the two methods in terms of residual level.

